Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

For a switched-affine plant following a prescribed fixed-period mode sequence, static cycle-to-cycle feedback can locally stabilize a reference periodic trajectory by shifting its $\boldsymbol{N-1}$ interior switching instants, but its raw switching-instant offsets may produce inadmissibly short or negative dwell durations. With strict nominal dwell margin, we multiply the complete raw vector by the largest conditioning factor in $\boldsymbol{[0,1]}$ that keeps every dwell duration admissible; the resulting applied switching-instant offsets preserve the requested direction, mode order, and cycle period. One scan of the $\boldsymbol{N}$ dwell changes computes the factor in $\boldsymbol{O(N)}$ time, with no online solver. In the linearized one-cycle model, the conditioned closed loop is $\boldsymbol{A(\beta) = (1-\beta)\Phi + \beta A_{\mathrm{cl}}}$, where $\boldsymbol{\Phi}$ is the nominal cycle matrix and $\boldsymbol{A_{\mathrm{cl}}}$ is the full raw-feedback matrix. One quadratic Lyapunov function that contracts both endpoints certifies arbitrary, including state-dependent, conditioning-factor sequences in that model. We derive the one-cycle linearization from exact augmented matrix-exponential propagation for mode-dependent affine dynamics. In an exact nonlinear 100-cycle simulation of a nine-interval DC–DC converter benchmark, an aggressive LQR raw action produces a schedule with a -43.045 µs dwell, and conditioning enforces the assumed 3 µs bound. The conditioned aggressive loop first crosses $\boldsymbol{\|S_x^{-1}e_k\|_2 < 0.01}$ at cycle 6, versus cycle 20 for the conservative LQR, which remained feasible throughout the run. The exact nonlinear guarantee is local to a neighborhood where conditioning is inactive; the large-error result is simulation evidence, not a global or experimental guarantee.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, one-cycle map, DC–DC converter

1 Introduction

Power converters set voltage and current by switching semiconductor devices in a repeating pattern. Their controllers work under hard limits: a fixed switching frequency, a minimum duration for every switch configuration, current and voltage bounds, and a short interval in which to decide. Predictive control enforces such limits explicitly, at the cost of an optimization problem solved online, and much recent work attacks that cost. Saeed et al. shrink the decision space with a Laguerre parameterization Saeed et al. (2022), Yang et al. build a solver specialized to fixed-frequency operation Yang et al. (2022), and others prune or approximate the candidate set Augustine and Patil (2023); Sakha and Kamalapurkar (2025). Stellato et al. (2017) combined horizons of one or two with an offline approximate tail cost and fixed-point exhaustive evaluation to obtain deterministic FPGA execution times. Static feedback removes the online optimization altogether. It does not remove the limits, and the action it requests can still be impossible to apply.

This paper works in one narrow setting. A prescribed sequence of affine modes repeats with a fixed period, and once per cycle the controller displaces only the interior switching instants. The mode order, the number of intervals, and the two cycle boundaries stay fixed, so the target is a periodic trajectory rather than an equilibrium. The actuator is therefore different from state-dependent mode selection, where the controller decides which mode is active Egidio et al. (2022); Russo et al. (2026), and from online schedule optimization, where a solver rebuilds the cycle at every step. It also differs from the hysteresis-band feedback of Repecho et al. (2017), which regulates measured switching periods by changing the band amplitude. Here one cycle-start error produces one static correction to the prescribed interior switching instants of the next cycle.

Two published results supply the cycle model. Patiño, Riedinger, and Ruiz derived switching-instant sensitivities for cyclic systems whose dynamics depend on the active mode and used them inside a predictive controller Patiño et al. (2010). Marcolino, Galvão, and Kienitz fixed the period and derived a one-cycle model, a discrete LQR timing law, and a dwell-constrained predictive controller for an LTI plant driven by switched actuator levels Marcolino et al. (2021). Their predictive controller enforces the dwell limits. Their standalone LQR comparison instead saturates the offsets one instant at a time, which can move the final switching instant, and the reported response diverges Marcolino et al. (2021). That example exposes the general difficulty. Each dwell duration depends on two adjacent offsets, so an unconstrained timing action can violate the limits, and clipping the offsets individually changes the direction the law asked for. Trading online optimization for static feedback does not by itself yield a schedule the converter can execute.

We close that gap without touching the feedback law. Given the raw offset vector, we apply the largest scalar multiple of it that keeps every dwell duration at or above the minimum. Scaling the whole vector by one number preserves its direction, the mode order, and both cycle boundaries, and one pass over the N dwell changes finds that number, so the online cost is $O(N)$ and no solver runs. In the linearized cycle model the conditioned closed loop is $A(\beta) = (1-\beta)\Phi + \beta A_{\mathrm{cl}}$, so a single quadratic Lyapunov function that contracts Φ and A_{cl} certifies every conditioning sequence the plant can generate. These two results are the contribution. Everything else supports them: the fixed-period one-cycle model for mode-dependent affine dynamics, which recovers the published sensitivity structure, Jacobians checked against finite differences, and an exact nonlinear simulation of a nine-interval converter benchmark. The LQR laws are comparison controllers, not a new synthesis method. The nonlinear guarantee is local, and the converter study is simulation rather than hardware validation. Section 2 derives the cycle model, section 3 builds the conditioner and its certificate, and section 4 reports the converter study.

2 Fixed-period one-cycle model

The conditioner of section 3 needs two things from the plant: how a switching-instant offset changes the next cycle-start error, and what dwell durations that offset produces. This section derives both, in four steps. We fix the timing coordinates, propagate one cycle exactly in an augmented state, expand that product to first order in the dwell changes, and rewrite the result in the independent switching-instant offsets.

2.1 Switched-affine cycle and timing coordinates

Consider

$$\dot{\boldsymbol{x}}(t) = A_{\sigma(t)}\boldsymbol{x}(t) + \boldsymbol{b}_{\sigma(t)}, \qquad \boldsymbol{x}(t) \in \mathbb{R}^n, \quad (1)$$

where each cycle follows the fixed mode sequence $\boldsymbol{\Omega} = (\omega_1, \ldots, \omega_N)$ with nominal boundaries

$$0 = \bar{t}_0 < \bar{t}_1 < \cdots < \bar{t}_N = T, \qquad \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \quad (2)$$

The nominal schedule generates a reference periodic trajectory $\bar{\boldsymbol{x}}(t)$. Its cycle anchor $\boldsymbol{x}^\star = \bar{\boldsymbol{x}}(0) = \bar{\boldsymbol{x}}(T)$ is a fixed point of the nominal cycle, and we measure the cycle-start error against it as $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}^\star$.

Only the interior switching instants move:

$$t_{k,i} = \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \\ \delta\tau_{k,0} = \delta\tau_{k,N} = 0. \tag{3}$$

The vector $\delta\boldsymbol{\tau}_k$ contains the $N-1$ independent offsets. The corresponding N dwell changes satisfy

$$\delta\boldsymbol{d}_k = D\delta\boldsymbol{\tau}_k, \qquad D_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j+1, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

for $i = 1, \dots, N$ and $j = 1, \dots, N-1$, so $\delta d_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}$. Moving one instant later lengthens the interval before it and shortens the interval after it by the same amount. Since $\mathbf{1}^\mathsf{T} D = 0$, the dwell changes sum to zero and every offset vector preserves the cycle period T. We write the exact one-cycle map as $\boldsymbol{x}_{k+1} = \mathcal{P}(\boldsymbol{x}_k, \delta\boldsymbol{\tau}_k)$.

2.2 Exact augmented propagation

Affine dynamics do not compose as a product of transition matrices, because each interval carries its own forced term. Appending a constant coordinate to the state removes that obstacle. Define the augmented state, the projection back onto the physical state, and the mode generators as

$$\mathcal{X} = \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix}, \qquad \Pi = \begin{bmatrix} I_n & \mathbf{0} \end{bmatrix}, \qquad F_i = \begin{bmatrix} A_{\omega_i} & \boldsymbol{b}_{\omega_i} \\ \mathbf{0}^\mathsf{T} & 0 \end{bmatrix}. \tag{5}$$

On interval i the augmented dynamics are linear, $\dot{\mathcal{X}} = F_i\mathcal{X}$. Write $E_i(s) = e^{F_i s}$ for a transition of duration s, and $\phi_i = E_i(\bar{d}_i)$ for the nominal one. The applied durations are $d_{k,i} = \bar{d}_i + \delta d_{k,i}$, so one cycle propagates exactly as

$$\mathcal{X}_{k+1} = E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1})\mathcal{X}_k. \tag{6}$$

Products act from right to left, so the rightmost factor covers the first interval. Nothing has been approximated yet. Let

$$\bar{\mathcal{X}}_i = \phi_i \cdots \phi_1 \bar{\mathcal{X}}_0, \qquad \widetilde{\boldsymbol{e}}_k = \begin{bmatrix} \boldsymbol{e}_k \\ 0 \end{bmatrix}. \tag{7}$$

Periodicity gives $\phi_N \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_0$, so subtracting the nominal cycle from (6) leaves the exact augmented terminal error

$$\widetilde{\boldsymbol{e}}_{k+1} = E_N(\bar{d}_N + \delta d_{k,N}) \cdots E_1(\bar{d}_1 + \delta d_{k,1}) \\ \times (\bar{\mathcal{X}}_0 + \widetilde{\boldsymbol{e}}_k) - \phi_N \cdots \phi_1 \bar{\mathcal{X}}_0. \tag{8}$$

2.3 First-order expansion in dwell durations

Only the interval durations vary in (8), and each one enters through a single matrix exponential. Because F_i is constant on interval i, differentiating with respect to that duration gives

$$\frac{\mathrm{d}}{\mathrm{d}s} e^{F_i s} = F_i e^{F_i s}, \tag{9}$$

and each perturbed transition satisfies

$$E_i(\bar{d}_i + \delta d_{k,i}) = \phi_i + F_i \phi_i \, \delta d_{k,i} + \mathcal{O}(\delta d_{k,i}^2). \tag{10}$$

Substituting (10) into the ordered product and keeping the terms with at most one dwell perturbation, with $E_i = E_i(\bar{d}_i + \delta d_{k,i})$, gives

$$E_N \cdots E_1 = \widetilde{\Phi} + \sum_{i=1}^{N} \phi_N \cdots \phi_{i+1} F_i \phi_i \cdots \phi_1 \, \delta d_{k,i} \\ + \mathcal{O}(\|\delta\boldsymbol{d}_k\|^2), \qquad \widetilde{\Phi} = \phi_N \cdots \phi_1, \tag{11}$$

where an empty product is the identity. The nominal term cancels when this expansion enters (8). Products of the form $\delta d_{k,i}\widetilde{\boldsymbol{e}}_k$ pair an error with a dwell change and are second order, so we drop them. Using $\phi_i \cdots \phi_1 \bar{\mathcal{X}}_0 = \bar{\mathcal{X}}_i$, the first-order terms that remain are

$$\widetilde{\boldsymbol{e}}_{k+1} = \widetilde{\Phi}\widetilde{\boldsymbol{e}}_k + \sum_{i=1}^{N} \widetilde{\boldsymbol{g}}_i \delta d_{k,i} + \mathcal{O}\left(\left\|[\boldsymbol{e}_k^\mathsf{T} \;\; \delta\boldsymbol{d}_k^\mathsf{T}]\right\|^2\right), \tag{12}$$

with

$$\widetilde{\boldsymbol{g}}_i = \phi_N \cdots \phi_{i+1} F_i \bar{\mathcal{X}}_i. \tag{13}$$

Projecting onto the physical state gives

$$\boldsymbol{e}_{k+1} = \Phi\boldsymbol{e}_k + G_d \delta\boldsymbol{d}_k + \mathcal{O}\left(\left\|[\boldsymbol{e}_k^\mathsf{T} \;\; \delta\boldsymbol{d}_k^\mathsf{T}]\right\|^2\right), \tag{14}$$

where

$$\Phi = e^{A_{\omega_N}\bar{d}_N} \cdots e^{A_{\omega_1}\bar{d}_1}, \qquad G_d = \Pi[\widetilde{\boldsymbol{g}}_1 \;\; \cdots \;\; \widetilde{\boldsymbol{g}}_N]. \tag{15}$$

2.4 Switching-instant offsets

Substituting $\delta \boldsymbol{d}_k = D\delta\boldsymbol{\tau}_k$ into (14) gives the model in the independent timing coordinates

$$\begin{aligned}\boldsymbol{e}_{k+1} &= \Phi\boldsymbol{e}_k + \Gamma_\tau\delta\boldsymbol{\tau}_k + \mathcal{O}\Big(\big\|[\boldsymbol{e}_k^\mathsf{T}\ \delta\boldsymbol{\tau}_k^\mathsf{T}]\big\|^2\Big), \\ \Gamma_\tau &= G_d D.\end{aligned} \tag{16}$$

Column i of D increases dwell i and decreases dwell $i+1$. Therefore $\Gamma_{\tau,i} = \Pi(\widetilde{\boldsymbol{g}}_i - \widetilde{\boldsymbol{g}}_{i+1})$. Since F_{i+1} commutes with $\phi_{i+1} = e^{F_{i+1}\bar{d}_{i+1}}$, factoring the adjacent terms yields

$$\begin{aligned}\Gamma_{\tau,i} &= e^{A_{\omega_N}\bar{d}_N}\cdots e^{A_{\omega_{i+1}}\bar{d}_{i+1}} \\ &\quad\times\big[(A_{\omega_i} - A_{\omega_{i+1}})\bar{\boldsymbol{x}}(\bar{t}_i) + \boldsymbol{b}_{\omega_i} - \boldsymbol{b}_{\omega_{i+1}}\big].\end{aligned} \tag{17}$$

The bracket is the jump in the affine vector field at the switching state, from mode ω_i to mode ω_{i+1}. That is what displacing instant i does. It trades time between two adjacent modes without changing the state at that instant to first order, and the remaining nominal transitions carry the resulting jump to the end of the cycle. Flieller et al. (2006) also used switching-time sensitivities to compute hybrid limit cycles and test their sampled-time local stability. Equation (17) therefore has the switching-sensitivity structure reported in Patiño et al. (2010), obtained here from dwell perturbations in independent fixed-period timing coordinates.

If every interval instead shares $A_{\omega_i} = A_c$ and $\boldsymbol{b}_{\omega_i} = B_c\boldsymbol{u}_{i-1}$, then

$$\Phi = e^{A_cT}, \qquad \Gamma_{\tau,i} = e^{A_c(T-\bar{t}_i)}B_c(\boldsymbol{u}_{i-1} - \boldsymbol{u}_i), \tag{18}$$

which matches Marcolino's switched-actuator timing model, apart from the sign convention for the input jump Marcolino et al. (2021). When A_{ω_i} depends on the mode, the jump also depends on the switching state, so this reduction does not apply in general.

3 Solver-free dwell-time conditioning

Briat and Seuret (2013) give stability tests for uncertain linear switched systems under minimum and mode-dependent dwell-time conditions. Here, section 2 says what an offset does, not whether it can be applied. A static law knows nothing about the minimum dwell duration, and (4) ties each dwell to two adjacent offsets, so clipping the offsets one at a time would change the direction of the correction. We keep the direction and give up magnitude instead, then certify the closed loop that results.

Let a static feedback law produce the raw switching-instant offset

$$\delta\boldsymbol{\tau}_k^{\mathrm{raw}} = -K\boldsymbol{e}_k, \qquad A_{\mathrm{cl}} = \Phi - \Gamma_\tau K. \tag{19}$$

Let $d_{\min}$ be the required minimum dwell duration and assume that the nominal schedule keeps a strict margin above it,

$$\boldsymbol{m} = \bar{\boldsymbol{d}} - d_{\min}\mathbf{1} > \mathbf{0}. \tag{20}$$

Each m_i is how much dwell i can be shortened before it becomes inadmissible. The raw action is feasible for exactly those errors in the polyhedron

$$\mathcal{C} = \{\boldsymbol{e} : \bar{\boldsymbol{d}} - DK\boldsymbol{e} \geq d_{\min}\mathbf{1}\}. \tag{21}$$

For $\boldsymbol{e}_k \in \mathcal{C}$ the raw action needs no modification. Outside $\mathcal{C}$ at least one dwell would fall below $d_{\min}$. Collect the requested dwell changes in $\boldsymbol{q}_k = D\delta\boldsymbol{\tau}_k^{\mathrm{raw}}$ and set

$$\begin{aligned}\beta_k &= \min\left\{1,\ \min_{i:q_{k,i}<0}\frac{m_i}{-q_{k,i}}\right\}, \\ \delta\boldsymbol{\tau}_k^{\mathrm{app}} &= \beta_k\delta\boldsymbol{\tau}_k^{\mathrm{raw}}.\end{aligned} \tag{22}$$

where a minimum over an empty set is $+\infty$. We call β_k the conditioning factor. It equals one whenever the raw action is already feasible, and otherwise the dwell that runs out of margin first decides it.

Proposition 1 (Feasibility and maximality)**.** *Under (20), (22) returns the largest $\beta \in [0,1]$ such that every applied dwell duration satisfies $\bar{\boldsymbol{d}} + D(\beta\delta\boldsymbol{\tau}_k^{\mathrm{raw}}) \geq d_{\min}\mathbf{1}$. The conditioned action preserves the cycle period and the direction of the raw action. Computing it takes $O(N)$ time and requires no online solver.*

Proof A dwell with $q_{k,i} \geq 0$ cannot be shortened by scaling. A dwell with $q_{k,i} < 0$ requires $\beta \leq m_i/(-q_{k,i})$. Intersecting these bounds with $[0,1]$ gives exactly (22), which is therefore both feasible and maximal. The identity $\mathbf{1}^\top D = 0$ preserves T, and multiplying the full action by one nonnegative scalar leaves its direction unchanged. A single pass over the N dwell changes evaluates every bound. □

In the linearized model, the applied action gives

$$\boldsymbol{e}_{k+1} = A(\beta_k)\boldsymbol{e}_k, \qquad A(\beta) = (1-\beta)\Phi + \beta A_{\mathrm{cl}}. \tag{23}$$

Conditioning therefore cannot produce an arbitrary saturated gain. It produces a matrix on the segment between the nominal cycle matrix Φ, obtained at $\beta = 0$ when the correction is switched off, and the raw closed-loop matrix A_{cl}, obtained at $\beta = 1$ when the correction is applied in full.

Proposition 2 (Common-Lyapunov contraction)**.** *Let $P = P^\top > 0$, $\|\boldsymbol{x}\|_P = (\boldsymbol{x}^\top P \boldsymbol{x})^{1/2}$, and suppose*

$$\|\Phi\|_P \le q_0 < 1, \qquad \|A_{\mathrm{cl}}\|_P \le q_1 < 1. \tag{24}$$

Then every $A(\beta)$ with $\beta \in [0,1]$ contracts in the same norm by a factor no greater than $q = \max(q_0, q_1) < 1$. The conditioned linearized model is therefore exponentially stable for every sequence $\{\beta_k\} \subset [0,1]$, including the state-dependent sequences generated by (22).

Proof The induced-norm triangle inequality gives $\|A(\beta)\|_P \le (1-\beta)q_0 + \beta q_1 \le q < 1$. Therefore $\|\boldsymbol{e}_k\|_P \le q^k \|\boldsymbol{e}_0\|_P$ for any sequence. □

The hypothesis on Φ deserves attention. It requires the uncontrolled cycle map to contract in the same norm, so the certificate is available only for plants whose nominal cycle is already stable. Without it the conditioner still runs and still returns an admissible schedule, and only the guarantee is lost. When such a P does exist, it can be computed offline from endpoint Lyapunov inequalities, which are standard semidefinite constraints Boyd et al. (1994). Online execution only evaluates the matrix–vector feedback law, $D\delta\boldsymbol{\tau}^{\mathrm{raw}}$, and the scalar scan in (22).

The nonlinear statement is weaker, and local. Suppose $\mathcal{P}$ is continuously differentiable while the mode order is preserved, and that A_{cl} is Schur. The strict nominal margin and $\delta\boldsymbol{\tau}^{\mathrm{raw}}(\mathbf{0}) = \mathbf{0}$ put a neighborhood of the anchor inside $\mathcal{C}$. Inside that neighborhood the conditioner never acts, $\beta = 1$, and the Jacobian of the exact closed-loop map is A_{cl}, so the exact map exponentially stabilizes the cycle anchor and the phase-locked reference periodic trajectory through it. Beyond that neighborhood, proposition 2 still covers arbitrary conditioning, but only for the linearized model. It does not prove global stability of the nonlinear plant.

4 Converter case study

The case study asks three questions of the theory. Does the derived one-cycle model reproduce the exact plant? Does the certificate of proposition 2 hold for a concrete design? And does conditioning deliver anything that an already-feasible controller cannot?

4.1 Three-cell converter benchmark and implementation

The hybrid-affine converter analysis of Albea-Sanchez et al. (2021) embeds PWM and sample-and-hold variables and uses exact two-subinterval matrix-exponential propagation in a Lyapunov analysis. The present case study uses a prescribed nine-interval mode sequence. We use the three-cell multilevel DC–DC converter studied by Patiño et al. Patiño et al. (2010), with state $\boldsymbol{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0,1\}$,

$$A_{\boldsymbol{u}} = \begin{bmatrix} 0 & 0 & (u_2-u_1)/C_1 \\ 0 & 0 & (u_3-u_2)/C_2 \\ (u_1-u_2)/L & (u_2-u_3)/L & -R/L \end{bmatrix}, \qquad \boldsymbol{b}_{\boldsymbol{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \tag{25}$$

Both $A_{\boldsymbol{u}}$ and $\boldsymbol{b}_{\boldsymbol{u}}$ depend on the switch state. The plant therefore does not satisfy the common-state-matrix assumption in (18). Table 1 collects the converter, schedule, and feedback parameters used in this section.

The reported physical mode sequence is $(0,1,3,7,2,0,4,7,4)$, the rounded cycle boundaries are $(0,66,88,110,132,154,220,242,264,286)$ µs, and the reported cycle anchor is $\boldsymbol{x}^\star = (9.9247, 19.2928, 0.9823)^\top$. Because those values are rounded, exact propagation through them does not return to the anchor. We therefore adjust the dwell

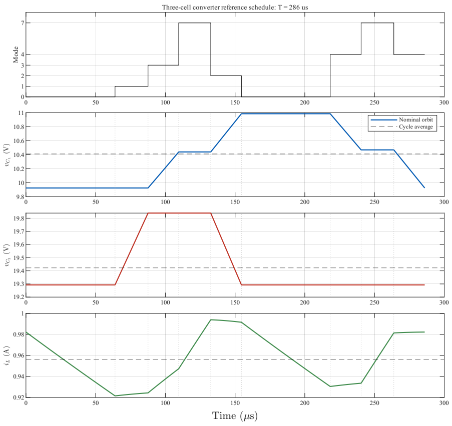

Figure 1 Reconciled reference schedule and exact periodic trajectory for the three-cell converter. Dashed horizontal lines show cycle averages. Dotted vertical lines mark the eight movable interior switching instants.

vector once by least squares, holding the published 286 µs period, the reported anchor, and the 22 µs nominal-design bound. The resulting schedule is

$$\bar{\boldsymbol{t}} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, \\ 218.210, 240.380, 263.828, 286)\ \mu\text{s}. \tag{26}$$

The largest boundary correction is 2.111 µs, and the closure error in the infinity norm is 1.421e-14. The adjustment prepares the benchmark data. It is not a trajectory-design method.

The cycle averages in figure 1 are 10.410 56 V, 19.422 28 V, and 0.956 13 A. We design in the normalized coordinates $\boldsymbol{z} = S_x^{-1}\boldsymbol{e}$ and $\boldsymbol{v} = \delta\boldsymbol{\tau}/t_s$, which bring the three states and the eight offsets to comparable scale. The aggressive raw law is a standard discrete LQR with $Q_n = I_3$ and $R_n = 0.001 I_8$. The conservative comparison law changes only R_n, to $0.1 I_8$. Only the aggressive law is conditioned.

4.2 Linearization and certificate checks

We check the analytical Jacobians against central differences of the exact one-cycle map. The state steps are 10^{-4} times the diagonal entries of S_x, and the timing step is $10^{-3} t_s = 0.01$ µs. A second check applies simultaneous state and timing perturbations with normalized amplitude ϵ and evaluates

$$r(\epsilon) = \left\| S_x^{-1} \left[\mathcal{P}(\boldsymbol{x}^\star + \Delta\boldsymbol{x}, \delta\boldsymbol{\tau}) - \boldsymbol{x}^\star \right.\right. \\ \left.\left. - \Phi\Delta\boldsymbol{x} - \Gamma_\tau \delta\boldsymbol{\tau} \right] \right\|_2. \tag{27}$$

The Jacobian errors and residual slope in table 2 confirm the implemented first-order model. The relative errors sit at the level of numerical noise, and the residual decays with slope 2.000, the second-order decay a correct first-order model must show. The generated data include the full analytical and finite-difference matrices and the residual sweep.

For the aggressive normalized closed loop, an offline semidefinite program returns a common P. The endpoint norms are $\|\Phi\|_P = 0.999939$ and $\|A_{\mathrm{cl}}\|_P = 0.979743$, so proposition 2 certifies the whole conditioned linearized family with contraction bound 0.999939. The open-loop endpoint sets that bound, because it is the slower of the two. The bound is therefore a worst case over arbitrary conditioning sequences, including the sequence that never applies the correction, and it does not describe the observed rate. Once β_k returns to one, the closed loop contracts at $\rho(A_{\mathrm{cl}}) = 0.568151$. The controller solves no semidefinite program online.

4.3 Large-error simulation

We simulate the exact nonlinear map for 100 cycles from

$$\boldsymbol{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^\mathsf{T}. \tag{28}$$

From this state we run the conditioned aggressive law, the unconditioned conservative law, and the open-loop schedule. Exact matrix exponentials propagate every interval, so the simulation never uses the linearized model. The initial normalized error is 0.983946, far outside the perturbation range used to check the local linearization.

Without conditioning, the aggressive law requests a minimum dwell of -43.045 µs, which would invert an interval and put two cycle boundaries out of order. Figure 2 shows that first cycle in detail: the raw schedule places one boundary before its predecessor, while the applied schedule keeps the order and clears the assumed bound. Figure 3 then tracks the error, the conditioning factor, the offsets, and the minimum dwell across the transient. The

Table 1 Converter, schedule, and feedback parameters. The applied bound of 3 µs is a simulation assumption, not the nominal benchmark-design bound.

Quantity	Symbol	Value
Source; capacitances	E; C_1, C_2	30 V; 40 µF
Inductance; resistance	L; R	10 mH; 10 Ω
Intervals; period	N; T	9; 286 µs
Nominal-design dwell bound	d_{bench}	22 µs
Applied-schedule dwell bound	$d_{\min}$	3 µs
State; timing scale	S_x; t_s	diag(10, 20, 1); 10 µs
Aggressive LQR weights	Q_n; R_n	I_3; $0.001 I_8$
Conservative LQR weights	Q_n; R_n	I_3; $0.1 I_8$

Table 2 Linearization checks, stability certificate, and exact nonlinear simulation results. Settling denotes the first cycle for which $\|S_x^{-1} e_k\|_2 < 0.01$.

Quantity	Value
Relative errors: Φ; Γ_τ	8.518e-12; 8.344e-11
Linearization residual slope	2.000
Spectral radii: open; conservative; aggressive	0.999850; 0.941468; 0.568151
Certified conditioned-family contraction bound	0.999939
Minimum conditioning factor; active cycles	0.293978; 3
Maximum raw; applied offset (µs)	65.217; 19.172
Minimum raw; applied dwell (µs)	-43.045; 3.000
Minimum conservative dwell (µs)	15.502
Settling cycle: conditioned; conservative	6; 20
Final errors: conditioned; conservative; open	1.746e-15; 3.113e-06; 0.298010

conditioner acts on 3 of the 100 cycles and drops to $\beta_{\min} = 0.293978$, cutting the largest offset from 65.217 to 19.172 µs. The minimum applied dwell is 3.000 µs, exactly the assumed bound, which is what maximality predicts. The conservative law needs no conditioning and holds a minimum dwell of 15.502 µs, but it reaches the 0.01 error threshold only at cycle 20, against cycle 6 for the conditioned aggressive law.

Over 100 cycles the open-loop response can look divergent in the state-space view. It is not. Under the fixed nominal schedule the open-loop cycle map has spectral radius $\rho(\Phi) = 0.999850 < 1$, so the cycle-start error does converge, only very slowly. It stays above the normalized-error threshold 0.01 until cycle 22009, whereas the conditioned response crosses that threshold at cycle 6. Figure 4 shows the initial transient and the long-horizon convergence together.

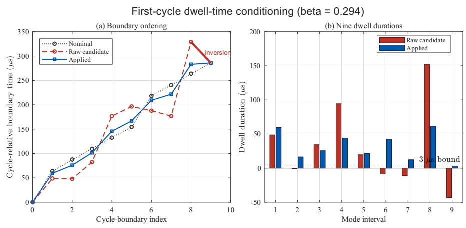

Figure 2 Dwell-time conditioning during the first cycle. (a) Nominal, raw, and applied cycle boundaries. The highlighted descending segment marks a boundary inversion in the raw schedule; conditioning preserves the boundary order. (b) Raw and applied durations for all nine dwells. The raw duration reaches -43.045 µs, while the applied schedule enforces the assumed 3 µs bound.

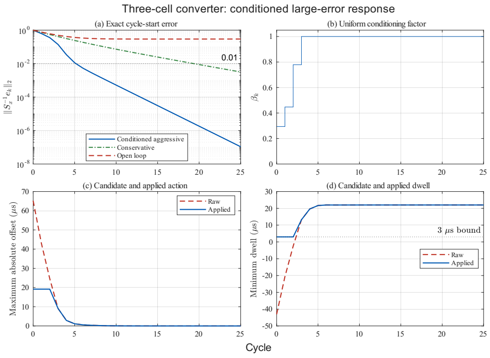

Figure 3 Exact nonlinear response from the large-error initial state, showing the first 25 of the 100 simulated cycles. (a) Normalized cycle-start errors with conditioned aggressive feedback, unconditioned conservative feedback, and open loop. (b) Conditioning factor. (c) Maximum absolute raw and applied switching-instant offsets. (d) Minimum raw and applied dwell durations. The raw duration becomes negative, while the applied schedule satisfies the assumed 3 μs bound.

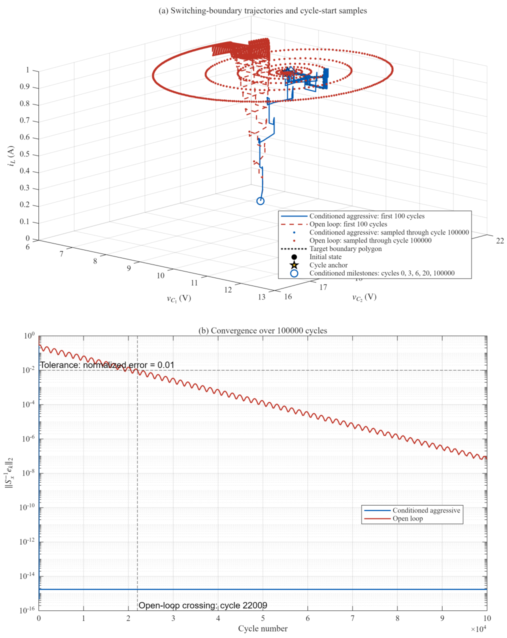

Figure 4 Exact large-error trajectory comparison over 100000 cycles. (a) The first 100 cycles are shown as exact switching-boundary paths; later points are cycle-start samples through cycle 100000 for both conditioned aggressive feedback and open loop. Markers identify conditioned cycle starts at cycles 0, 3, 6, 20, and 100000. (b) Normalized cycle-start errors over the common horizon. The open-loop response converges to the target switching limit cycle but remains above 0.01 until cycle 22009, whereas the conditioned response crosses it at cycle 6. These simulations are benchmark evidence and do not establish global nonlinear stability.

5 Discussion

The conditioner keeps the largest feasible scalar multiple of a static timing action and changes neither its direction nor the cycle boundaries. That is also its limit. It cannot look ahead, and it cannot choose a different direction when the requested one is nearly blocked, so it does not replace constrained optimization for controllers that must do either. The search for a common P runs offline and costs nothing online.

The cycle model joins two earlier formulations. Patiño et al. derived the mode-dependent switching sensitivity Patiño et al. (2010), and Marcolino et al. formulated fixed-period timing coordinates, dwell inequalities, and LQR for switched actuators Marcolino et al. (2021). Our derivation connects the two and checks the result numerically. The new control result is the scalar conditioner and its common-Lyapunov certificate, not the LQR design.

The 3 µs minimum dwell is a simulation assumption rather than a hardware figure. The study leaves out timer quantization, delays, dead time, implementation margin, state estimation, model uncertainty, and experiments, and it fixes the mode order, phase, and period. The exact nonlinear guarantee holds only in a neighborhood where conditioning never acts, so the large-error simulation is evidence for this benchmark and not a global nonlinear result.

6 Conclusion

The conditioner enforces every minimum-dwell constraint in $O(N)$ time and keeps both the direction of the raw action and the cycle period. One quadratic Lyapunov function for the two endpoint matrices certifies the entire conditioned linearized family, including the state-dependent conditioning sequences the plant generates. The augmented derivation behind those matrices covers mode-dependent switched-affine dynamics and reduces to the switched-actuator model when the state matrix is common, and finite differences confirm the Jacobians it produces. On the converter benchmark, conditioning turns an infeasible aggressive action into an admissible one, and the response still settles sooner than that of a conservative law chosen to be feasible throughout. The nonlinear guarantee remains local, and hardware tests are still needed to account for timer quantization, delays, and implementation margin.

Declarations

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Data availability. TODO: state where the generated simulation data are available, or that all data supporting the findings are contained in the article. Required for original research.

Code availability. TODO: state where the simulation and figure-generation scripts are available, or that this item is not applicable.

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